

Software Requirements Specification

Kesuma Dewi (KD)

AI Chatbot for Kementerian Sumber Manusia (KESUMA / MOHR)

Version 1.0 · 3 June 2026

Build specification for AI-assisted development

Stack: Laravel · Vue 3 · MySQL · Corrad-Laravel base

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Prepared as: Build specification for AI-assisted ("vibe coding") development

Target stack: Laravel (API) · Vue 3 (SPA/admin) · MySQL · Corrad-Laravel base (<https://github.com/mfauzzury/corrad-laravel>)

How to use this document

This SRS is written so that a developer using AI-assisted ("vibe") coding can read it top-to-bottom and build the complete web application without further clarification. It is organised as:

1. Context and goals (Sections 1–3)
2. The two product segments — **Public** and **Admin** (Sections 4–5)
3. Cross-cutting capabilities — AI/RAG, knowledge base, crawler, notifications (Sections 6–9)
4. Non-functional requirements, data model, APIs, and acceptance criteria (Sections 10–15)

Every functional requirement is tagged (e.g. **FR-PUB-03**) so it can be referenced in tickets, prompts, and tests. Where a requirement maps to a feature already present in the Corrad-Laravel base repo, it is noted as **[Corrad: reuse]**.

1. Introduction

1.1 Purpose

The purpose of this system is to provide a multilingual, AI-powered (not rule-based) conversational assistant — named Kesuma Dewi, or KD — that serves the Malaysian public on behalf of Kementerian Sumber Manusia (KESUMA / MOHR) and its agencies. KD answers questions, guides users to the right service, and helps with enquiries focused (initially) on three agency systems, while giving MOHR and agency administrators a back office to manage knowledge, conversations, and users.

1.2 The name "Kesuma Dewi (KD)"

- **Kesuma** — derived from **KESUMA**, the Malay short-form of the Ministry (KEmenterian SUMber MAnusia).
- **Dewi** — used as a friendly persona name and as a backronym expressing the AI nature of the assistant. Recommended expansion: **D**igital **E**nquiry & **W**eb **I**ntelligence (alternatives the team may pick from: "Digital Engagement & Workforce Intelligence", "Data-driven Enquiry & Web Intelligence"). The chosen expansion must be configurable as a setting (see FR-SET-02), not hard-coded.
- **Persona avatar**: A friendly young Malay woman wearing a *tudung* (headscarf) and traditional *kebaya*, presented as a small floating chat icon on the public website. The avatar asset is a configurable image so MOHR can replace it without code changes.

1.3 Scope

In scope

- A public-facing replica of the MOHR front page (<https://www.mohr.gov.my/>) for prototype purposes, with the KD chat widget embedded.
- A conversational onboarding and Q&A flow for the public (language selection, introduction, identity capture with verification, agency-focused assistance).
- An admin back office with role-based access control (RBAC), knowledge-base management, conversation/chat management, agency administration, dashboards, settings, and an automated web crawler that feeds the knowledge base.
- AI answering grounded in the knowledge base (Retrieval-Augmented Generation), with conversation transcripts fed back into the knowledge base.

Out of scope (for this phase)

- Native mobile apps (the web app must be responsive instead).
- Direct write-transactions into agency core systems (KD reads/looks up and guides; it does not file official claims/complaints on the agency's behalf unless a future integration is approved).
- Migrating MOHR's existing Joomla portal; only a front-page replica is produced.

1.4 Focus agencies (initial three)

KD initially focuses on three agencies/systems. The platform must allow adding more agencies later — knowledge and administration are organised per agency.

#	Agency	System / Service	Typical user intent
1	HRDCorp	MyFutureJob (Portal Pekerjaan Negara)	Job search queries, job availability data
2	JTK (Jabatan Tenaga Kerja)	Sistem Aduan Pekerja	Labour complaints & issues, complaint reference number lookups
3	PERKESO	Claim System	Claim applications, claim reference number lookups

1.5 Definitions, acronyms, abbreviations

- **MOHR / KESUMA** — Ministry of Human Resources / Kementerian Sumber Manusia.
- **KD** — Kesuma Dewi, the chatbot persona.
- **RBAC** — Role-Based Access Control.
- **KB** — Knowledge Base.
- **RAG** — Retrieval-Augmented Generation: AI answers grounded in retrieved KB content.
- **LLM** — Large Language Model (the AI engine answering questions).
- **Embedding** — a numeric vector representation of text used for semantic search.
- **SPA** — Single Page Application (the Vue 3 front end).
- **OTP** — One-Time Password (used for email/phone verification).
- **BM / EN / ZH / TA** — Bahasa Melayu / English / Chinese (Mandarin) / Tamil.

1.6 References

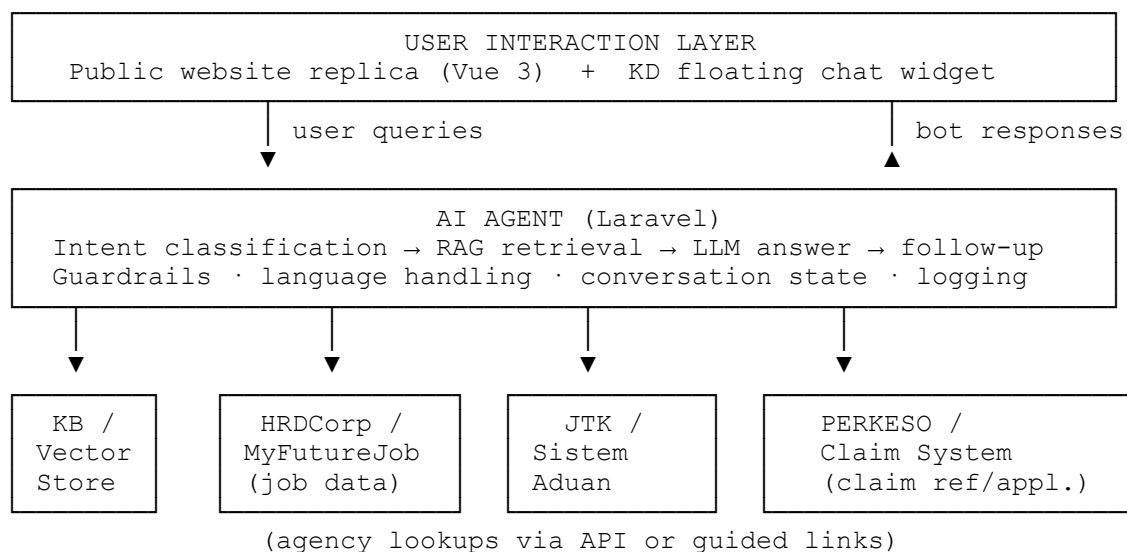
- MOHR portal: <https://www.mohr.gov.my/>
- Base repository / starter stack: <https://github.com/mfauzzury/corrad-laravel> (Laravel API + Vue 3 admin + Eloquent ORM + Tailwind CSS, with built-in content editing, media management, RBAC, menu management, dynamic site settings).

2. Overall description

2.1 Product perspective

KD is a new, cloud-hosted web application. It is not a rule-based decision tree; answering is driven by an LLM grounded in a curated, per-agency knowledge base (RAG). The system reuses the Corrad-Laravel base for authentication, RBAC, media management, settings, and the Vue 3 admin shell, and extends it with chatbot-specific modules.

High-level architecture (logical):



2.2 User classes (RBAC) — three categories

5. **Public user** — anonymous/lightly-identified end users chatting with KD on the public site. May provide name/email/phone (verified) but do not log in to an admin panel.
6. **Admin** — MOHR staff and **agency** staff who log in to the back office. Agency admins are scoped to their own agency's knowledge and conversations. Includes the ability for an agency admin to add additional admins for their own agency.
7. **Super Admin** — full platform control: manages all users, all agencies, global settings, crawler configuration, and system-wide knowledge base.

2.3 Operating environment

- **Cloud-hosted** (e.g. a Malaysian-region cloud or government cloud / MyGovCloud-compatible). Server-side Laravel app + MySQL database + a vector store + object storage for uploaded documents.
- **Client:** modern browsers (Chrome, Firefox, Edge, Safari), responsive for mobile and desktop.

2.4 Design & implementation constraints

- **Stack is fixed:** Laravel (PHP) for the API/back end, Vue 3 + Tailwind for the front end, MySQL as the primary relational database, building on the Corrad-Laravel base repo.
- **AI-based, not rule-based.** Answers come from an LLM grounded in the KB. Hard-coded decision trees are only acceptable for the deterministic onboarding steps (language pick, identity capture).
- **Bahasa Melayu first.** All public-facing copy must be available in BM, EN, ZH, TA.
- **Government data protection.** Personal data (PDPA Malaysia) must be handled per Section 10.4.

2.5 Assumptions & dependencies

- An LLM provider/API is available (cloud-hosted commercial model or a self-hosted open model). The provider is configurable (Section 9).
- Email and SMS/WhatsApp gateways are available for verification and notifications.
- Where agency live-lookups (e.g. claim/complaint reference status) require official APIs, those APIs are provided by HRDCorp/JTK/PERKESO; until then KD guides users to the correct agency portal and explains the process from the KB.

3. System goals & success criteria

- **G1** — A member of the public can get accurate, source-grounded answers about MOHR and the three focus agencies in their preferred language (BM/EN/ZH/TA).
- **G2** — KD greets users, introduces itself, captures and verifies identity politely, and routes enquiries to the right agency.
- **G3** — MOHR and agency admins can manage knowledge per agency, monitor and respond to conversations, and view dashboards.
- **G4** — The KB stays current automatically via scheduled crawling of MOHR/agency public content and by ingesting past conversations.
- **G5** — The platform is extensible to more than three agencies without code changes to the data model.

4. Public segment — functional requirements

4.1 Public website replica

- **FR-PUB-01** — The system shall present a **replica of the MOHR front page** (<https://www.mohr.gov.my/>) for prototype purposes, reproducing the front-page content and layout: top utility bar (Soalan Lazim, Hubungi Kami, Maklum Balas, Peta Laman), the MOHR logo/header, the main navigation (LAMAN UTAMA, INFO KORPORAT, SUMBER, MEDIA, LAMAN MIKRO, PERTANYAAN), the hero/banner carousel, "Perkhidmatan Dalam Talian" quick-link tiles (MyKKP, MyFutureJobs, ILJTM, ePPAx, TiPS, e-Industrial Court, eTraining HRDCorp, PTPK, MySPIKE), the "Info Lanjut" tabs (Kenyataan Media, Aktiviti Terkini, Tender), the "Pautan Pantas" panels, the "Jabatan & Agensi" logo grid, and the footer (address: Aras 6–12, Menara PERKESO Putrajaya, Presint 2, 62100 Putrajaya; Tel 03-8880 6200; e-mel aduan.ukk@mohr.gov.my; privacy/security/copyright/disclaimer links).
- **FR-PUB-02** — The replica shall be built in Vue 3 + Tailwind, be responsive, and store its front-page content as editable records (so admins can update banners, tiles, and links via the back office — **[Corrad: reuse content editing & media management]**). External quick-links shall point to the real agency URLs.
- **FR-PUB-03** — The replica is a *prototype shell*; it must not attempt to proxy or scrape live MOHR pages at render time. Front-page content is seeded once and thereafter editable in the admin.

4.2 The KD chat widget

- **FR-PUB-04** — A small **floating chat icon** (the Kesuma Dewi avatar — young Malay woman in tudung and traditional kebaya) shall appear on every public page, anchored bottom-right, with an unobtrusive "Tanya KD / Ask KD" label and a subtle attention animation on first visit.
- **FR-PUB-05** — Clicking the icon opens a chat panel with KD's avatar, name, a short tagline, the conversation thread, a text input, a send button, a language switcher, and a "restart conversation" control. The panel must be keyboard-accessible and screen-reader friendly (Section 10.5).
- **FR-PUB-06** — The avatar image, name, tagline, accent colour, and welcome copy shall be **configurable via Settings** (Section 8), not hard-coded.

4.3 Conversation onboarding flow

The opening flow is deterministic (scripted) before handing over to AI answering:

- **FR-PUB-07 — Language selection.** On first open, KD shall ask the user to choose a language: **Bahasa Melayu, English, 中文 (Chinese), தமிழ் (Tamil)**. The whole conversation (KD's messages, buttons, prompts) then continues in the selected language. The user may switch language at any time.
- **FR-PUB-08 — Self-introduction.** After language selection, KD shall introduce itself: who KD is (the official AI assistant of Kementerian Sumber Manusia), what it can help with, and a privacy note that the chat may be recorded and used to improve service (with a link to the privacy notice). Intro copy is per-language and editable in Settings.
- **FR-PUB-09 — Mention the three focus agencies.** During or right after the introduction, KD shall state that it can specifically help with the three focus services: **HRDCorp / MyFutureJob (jobs), JTK / Sistem Aduan Pekerja (labour complaints), and PERKESO / Claim System (claims)** —

and invite the user to pick one or type their question freely. Presented as quick-reply chips plus free text.

- **FR-PUB-10 — Capture name.** KD shall politely ask for the user's name so it can address them properly, and thereafter use the name naturally (e.g. "Baik, Encik/Puan {name}..."). Name capture is requested but skippable; if skipped, KD uses a neutral, polite address.
- **FR-PUB-11 — Capture & verify email.** KD shall ask for an email address and verify it by sending a 6-digit OTP (or a verification link) to that address; the user enters the code in chat to confirm. Unverified emails are flagged as such. Validation of format precedes sending.
- **FR-PUB-12 — Capture & verify phone.** KD shall ask for a mobile number and verify it via OTP sent by SMS or WhatsApp; the user enters the code to confirm. Phone capture/verification is requested; the user may decline and still continue with limited follow-up options.
- **FR-PUB-13 — Consent.** Before storing personal data, KD shall obtain explicit consent (a clear yes/no with a privacy-notice link). Without consent, the chat continues anonymously and no PII is stored.
- **FR-PUB-14 — Graceful skipping.** Every identity step (name, email, phone) must be skippable; KD must never block answering a general question because identity is incomplete, but may require a verified contact before promising an email/WhatsApp follow-up from a human agent.

4.4 The conversation itself

- **FR-PUB-15 — AI answering (RAG).** After onboarding, KD shall answer questions conversationally using the LLM grounded in the per-agency knowledge base (Section 6). Answers must be in the user's selected language regardless of the KB source language.
- **FR-PUB-16 — Intent classification.** Each incoming message shall be classified (e.g. job-search, labour-complaint, claim, general-MOHR, complaint-reference-lookup, claim-reference-lookup, out-of-scope) to select the right KB scope/agency and the right follow-up actions. Classification labels are stored on the message for analytics.
- **FR-PUB-17 — Agency routing & lookups.** Based on intent, KD shall (a) answer from the agency KB, (b) deep-link the user to the correct agency portal/form, and (c) where an official agency lookup API is configured, accept a reference number (e.g. complaint or claim reference) and return status. Until lookup APIs exist, KD explains the process and links the portal.
- **FR-PUB-18 — Grounding & citations.** KD's answers shall be grounded in retrieved KB passages; when confidence is low or no relevant KB content is found, KD must say so honestly, avoid fabricating, and offer to connect the user to a human (email/WhatsApp) or to file feedback. Optionally show a "source" reference for transparency.
- **FR-PUB-19 — Follow-up & escalation.** KD shall offer follow-up actions: continue chatting, get a transcript by email, or request a human agent. If a human is requested and a verified contact exists, the conversation is queued for admin response via email or WhatsApp (Section 7).
- **FR-PUB-20 — Conversation memory.** Within a session, KD shall retain context (name, language, prior turns) to answer coherently. A session can be resumed via a returning link if a verified email is on file (optional, configurable).
- **FR-PUB-21 — Tone & safety.** KD shall be warm, respectful, and formal-friendly in Malay government style; refuse abusive, illegal, or out-of-scope requests politely; and never give legal/financial determinations — only official information and guidance.
- **FR-PUB-22 — Feedback.** After an answer, the user may rate it (helpful / not helpful) and optionally leave a comment; ratings are stored for quality analytics.
- **FR-PUB-23 — Transcript storage.** Every public conversation shall be stored (Section 6.5) and, subject to consent and PII-handling rules, fed back into the knowledge base for future answering (FR-KB-09).

5. Admin segment — functional requirements

5.1 Authentication & RBAC

- **FR-ADM-01 — Login.** Admin and Super Admin users shall log in via secure authentication (email + password, with optional 2FA) [**Corrad: reuse auth**].
- **FR-ADM-02 — RBAC.** The system shall enforce role-based access control with the three categories **Public, Admin (MOHR & agencies), Super Admin**, plus fine-grained permissions [**Corrad: reuse RBAC**]. Permissions are grouped by module (users, KB, chat, crawler, settings, dashboard) and by agency scope.
- **FR-ADM-03 — Agency scoping.** An Admin tied to an agency shall only see and manage data (KB, conversations, sub-admins) belonging to that agency. MOHR-level Admins may be granted cross-agency visibility. Super Admin sees everything.
- **FR-ADM-04 — Agency admin self-service.** An agency Admin shall be able to **add additional admins for their own agency** (invite by email, assign agency-scoped role), subject to a per-agency seat/permission policy set by Super Admin.

5.2 User management

- **FR-ADM-05 — User management with RBAC.** Super Admin (and permitted MOHR Admins) shall create, view, edit, deactivate, and delete users; assign roles and agency scope; reset passwords; and view audit history [**Corrad: reuse user management**].
- **FR-ADM-06 — Three user categories.** The user model shall support the categories Public, Admin (MOHR/agency), and Super Admin, with agency association for agency admins. (Public chat users are stored separately from admin accounts — see data model 11.)
- **FR-ADM-07 — Audit log.** All privileged actions (user changes, KB edits, crawler config, settings changes, chat responses sent) shall be recorded in an immutable audit log with actor, timestamp, and before/after where relevant.

5.3 Knowledge base management

- **FR-ADM-08 — KB per agency.** The knowledge base shall be organised **by agency**. The system ships with the three focus agencies and allows Super Admin to **add more agencies**; each agency has its own KB collection and access scope.
- **FR-ADM-09 — Upload all knowledge types.** Admins shall upload knowledge of any common type — user manuals, website content, PDFs, Word/Excel/PowerPoint, images with text, plain text, FAQs, and URLs — tagged to an agency and topic [**Corrad: reuse media management**].
- **FR-ADM-10 — Ingestion pipeline.** On upload/import, the system shall extract text, chunk it, generate embeddings, and store vectors with metadata (agency, source, title, language, version, timestamp) so the content becomes retrievable by KD (Section 6).
- **FR-ADM-11 — KB lifecycle.** Admins shall edit metadata, re-index, version, archive, and delete KB items; deletions remove the corresponding vectors. Agency admins act only within their agency's KB.
- **FR-ADM-12 — Review of conversation-derived knowledge.** Knowledge generated from past conversations (FR-KB-09) shall enter a review queue; an admin approves, edits, or rejects before it becomes authoritative KB content.

5.4 Chat management

- **FR-ADM-13 — Conversation inbox.** Admins shall see a list/inbox of conversations they are permitted to view (agency-scoped), with filters (date, language, agency, intent, rating, status: open/awaiting-human/closed) and full-text search.
- **FR-ADM-14 — View logs.** Admins shall open any permitted conversation and view the complete transcript, captured identity (name, verified email/phone), language, intent labels, KB sources used, and metadata/timestamps.

- **FR-ADM-15 — Respond via email or WhatsApp.** From a conversation, an admin shall be able to respond to the user **by email or WhatsApp** (where a verified contact exists), with the outbound message recorded back into the transcript and audit log (Section 7).
- **FR-ADM-16 — Assignment & status.** Conversations needing human attention shall be assignable to an admin, with status transitions and SLA timestamps; agency admins handle their own agency's queue.
- **FR-ADM-17 — Export.** Admins shall export transcripts (PDF/CSV) subject to permission and PII rules.

5.5 Agency administration

- **FR-ADM-18 — Agency records.** Super Admin shall create/edit agencies (name, code, logo, contact, focus systems, KB collection, deep-link URLs, lookup-API config). The three focus agencies (HRDCorp/MyFutureJob, JTK/Sistem Aduan Pekerja, PERKESO/Claim System) are seeded.
- **FR-ADM-19 — Agency login & visibility.** Agency admins shall log in and see (only) their agency's chats and KB, and may add their own sub-admins (FR-ADM-04).

5.6 Dashboards

- **FR-ADM-20 — Role-aware dashboard.** On login, both MOHR and agency admins shall land on a **dashboard** scoped to their permissions, showing at minimum: total conversations (today/7d/30d), conversations by language, by agency, by intent, average rating/helpfulness, % answered by AI vs escalated to human, open/awaiting-human counts, top unanswered questions (gaps in KB), and crawler health/last-run status.
- **FR-ADM-21 — Drill-down.** Dashboard cards shall link to filtered views (e.g. click "awaiting human" → the relevant inbox filter).
- **FR-ADM-22 — Live artifact option.** Dashboards should support a shareable, auto-refreshing view for monitoring.

5.7 Crawler module

- **FR-ADM-23 — Crawl public content into KB.** The system shall crawl the public MOHR website and configured agency sites and store the extracted content in the knowledge base, tagged to the relevant agency.
- **FR-ADM-24 — Configure the crawl.** Admins shall configure crawl jobs: seed URLs, include/exclude URL patterns, crawl depth, max pages, file types to ingest (HTML/PDF/DOCX), agency tag, and language handling.
- **FR-ADM-25 — Schedule & interval.** Admins shall set the **schedule/time interval** for each crawl job (e.g. daily at 02:00, weekly, monthly, or a custom cron), enable/disable jobs, and trigger an on-demand run.
- **FR-ADM-26 — Incremental & dedup.** Crawls shall detect changed pages (via content hash/last-modified), update only changed content, avoid duplicates, and respect robots.txt and a politeness delay.
- **FR-ADM-27 — Crawl review & logs.** Each run shall produce a log (pages fetched, added, updated, skipped, errors). New crawled content may go to the KB directly or to a review queue per a per-job setting.

5.8 Settings module

- **FR-SET-01 — Central settings.** A settings module shall let Super Admin (and permitted Admins) configure the platform without code changes [**Corrad: reuse dynamic site settings**]. Proposed settings groups:
 - **Branding & persona** (FR-SET-02): KD avatar image, name, "Dewi" backronym expansion, tagline, accent colour, welcome/intro copy per language.
 - **Languages:** enabled languages (BM/EN/ZH/TA), default language, per-language UI strings.
 - **Onboarding:** which identity steps are required vs optional (name/email/phone), consent text, privacy-notice link.
 - **AI engine:** LLM provider/model, API keys (secret), temperature, max tokens, system prompt/ persona instructions, retrieval top-k, confidence threshold for "I don't know / escalate", and guardrail rules.
 - **Knowledge base:** chunk size/overlap, embedding model, re-index controls, default review-queue on/off.
 - **Crawler:** global politeness delay, default schedule, user-agent string.
 - **Notifications:** email (SMTP/API) and WhatsApp/SMS gateway credentials, OTP length & expiry, message templates per language.
 - **Escalation/SLA:** response-time targets, routing rules, business hours, auto-reply when offline.
 - **Security:** password policy, 2FA toggle, session timeout, data-retention period for transcripts/ PII.
 - **Agencies:** manage agency list and per-agency deep-links and lookup-API endpoints.
- **FR-SET-03 — Audit of setting changes.** All settings changes shall be audited (FR-ADM-07).

6. Knowledge base & AI answering (RAG)

- **FR-KB-01** — The KB shall store source documents, extracted text chunks, embeddings (vectors), and metadata (agency, title, source URL/file, language, version, created/updated, status).
- **FR-KB-02** — Supported ingestion sources: manual uploads (FR-ADM-09), crawler output (FR-ADM-23), and approved conversation-derived knowledge (FR-KB-09).
- **FR-KB-03** — Text extraction shall handle HTML, PDF, DOCX, XLSX, PPTX, TXT, and images via OCR where needed.
- **FR-KB-04** — Chunking shall split text into retrieval-sized passages with configurable size/overlap, preserving source metadata on each chunk.
- **FR-KB-05** — Embeddings shall be generated by a configurable embedding model and stored in a vector index. The vector store may be a MySQL-compatible vector solution or an external vector database; retrieval is by semantic similarity filtered by agency scope.
- **FR-KB-06 — Retrieval.** At query time the system shall retrieve the top-k most relevant chunks (optionally hybrid keyword + semantic), filtered to the relevant agency/scope, and pass them as grounding context to the LLM.
- **FR-KB-07 — Generation.** The LLM shall produce an answer using only the retrieved context plus its instructions/persona, in the user's language, citing or referencing sources where configured, and explicitly deferring when context is insufficient.
- **FR-KB-08 — Multilingual.** The KB may hold content in BM/EN; KD must answer in the user's chosen language (BM/EN/ZH/TA), translating as needed while preserving official terminology.
- **FR-KB-09 — Conversations back into the KB.** All conversations shall be captured and, after PII redaction and admin review (FR-ADM-12), eligible to become KB content (e.g. as approved Q&A pairs) so KD improves over time.
- **FR-KB-10 — Freshness.** Crawled content shall carry a "last-verified" timestamp; admins can prioritise official/verified sources over conversation-derived content during retrieval.

7. Notifications, email & WhatsApp

- **FR-NOT-01** — The system shall send transactional messages: email/SMS/WhatsApp OTPs for verification (FR-PUB-11/12), optional transcript-by-email, and human-agent replies (FR-ADM-15).
- **FR-NOT-02** — Email shall be sent via configurable SMTP or an email API; WhatsApp/SMS via a configurable gateway (e.g. WhatsApp Business API provider). Credentials live in Settings (FR-SET-01).
- **FR-NOT-03** — All templates shall be editable per language with variables (name, agency, reference number, link).
- **FR-NOT-04** — Outbound human replies shall thread back into the conversation record and the audit log.
- **FR-NOT-05** — OTPs shall have configurable length and expiry, with rate-limiting and lockout after repeated failures.

8. Branding, persona & localisation

- **FR-BRD-01** — KD's persona (avatar, name, tagline, voice/tone) shall be centrally configurable (Section 5.8).
- **FR-BRD-02** — Default persona: Kesuma Dewi — friendly, respectful young Malay woman in tudung and traditional kebaya; warm, formal-friendly Malaysian government tone.
- **FR-BRD-03** — All public strings shall be localisable to BM, EN, ZH, TA via a translation table; BM is the default.
- **FR-BRD-04** — The system prompt/persona instruction given to the LLM shall enforce identity ("You are Kesuma Dewi, the official AI assistant of Kementerian Sumber Manusia..."), scope, tone, honesty/no-fabrication, and safety rules; it is editable in Settings.

9. AI engine requirements

- **FR-AI-01** — Answering shall be **AI/LLM-based, not rule-based**; the onboarding steps may be scripted.
- **FR-AI-02** — The LLM provider and model shall be configurable (commercial API or self-hosted open model) with secret key storage.
- **FR-AI-03** — The engine shall support: system/persona prompt, retrieved-context injection (RAG), conversation history, language control, temperature/token settings, and a confidence/uncertainty threshold that triggers honest deferral or human escalation.
- **FR-AI-04** — Guardrails shall block out-of-scope, abusive, or unsafe content and prevent the model from inventing official facts, figures, legal advice, or claim/complaint outcomes.
- **FR-AI-05** — All AI calls shall be logged (prompt, retrieved sources, response, latency, token usage) for quality and cost monitoring, with PII handled per Section 10.4.

10. Non-functional requirements

10.1 Performance

- **NFR-01** — Median chatbot response latency ≤ 3 s and 95th-percentile ≤ 8 s under normal load (excluding third-party LLM outages).
- **NFR-02** — The system shall support at least 500 concurrent public chat sessions in the initial deployment, horizontally scalable thereafter.

10.2 Availability & reliability

- **NFR-03** — Target availability 99.5% for the public chat. Graceful degradation: if the LLM is unavailable, KD apologises, offers FAQ links and human escalation.
- **NFR-04** — Long-running work (crawling, ingestion, embeddings, notifications) shall run on asynchronous queues/workers, retried on failure.

10.3 Scalability & extensibility

- **NFR-05** — Adding a new agency shall require only configuration/data (new agency record + KB collection + admins), not schema or code changes.
- **NFR-06** — The crawler, KB, and notification subsystems shall be modular and independently scalable.

10.4 Security, privacy & compliance

- **NFR-07** — Comply with Malaysian PDPA and government data-protection expectations for handling personal data (name, email, phone, transcripts).
- **NFR-08** — Encrypt data in transit (TLS) and at rest (PII fields and stored documents).
- **NFR-09** — Explicit consent before storing PII; configurable data-retention and a deletion/erasure path on request.
- **NFR-10** — Secrets (LLM/email/WhatsApp keys) stored securely (env/secret manager), never exposed to the client.
- **NFR-11** — Enforce RBAC and agency scoping on every API endpoint server-side (never trust the client).
- **NFR-12** — Rate-limit public chat and OTP endpoints; protect against prompt-injection, spam, and abuse; sanitise crawled/ingested content.
- **NFR-13** — Maintain audit logs (FR-ADM-07) and redact PII before conversation content enters the KB.

10.5 Accessibility & UX

- **NFR-14** — Public site and chat widget shall meet WCAG 2.1 AA: keyboard navigation, screen-reader labels, sufficient contrast, scalable text.
- **NFR-15** — Responsive on mobile and desktop; the chat widget must be usable one-handed on a phone.

10.6 Maintainability & deployability

- **NFR-16** — Code shall follow the Corrad-Laravel conventions (Laravel API + Vue 3 SPA + Tailwind), with environment-based config, migrations, and seeders.
- **NFR-17** — Provide automated tests for critical paths (onboarding, OTP, RAG answering, RBAC enforcement, crawler scheduling) and a CI pipeline.
- **NFR-18** — Provide seed data for the three focus agencies and a demo super-admin.

11. Data model (logical)

Primary entities (MySQL via Eloquent; vectors in the configured vector store). Names are indicative.

- **users** — id, name, email, password_hash, phone, role_id, agency_id (nullable), is_active, 2fa, timestamps. Roles: super_admin, admin, (public users tracked separately).
- **roles / permissions / role_permission** — RBAC [**Corrad: reuse**].
- **agencies** — id, name, code, logo, contact, focus_system, portal_url, deep_links (json), lookup_api_config (json), is_active, timestamps.
- **public_users** (chat contacts) — id, display_name, email, email_verified_at, phone, phone_verified_at, preferred_language, consent_at, created_at. (Lightweight, separate from admin users.)
- **conversations** — id, public_user_id (nullable), agency_id (nullable), language, status (open/awaiting_human/closed), assigned_admin_id (nullable), rating, started_at, last_activity_at.
- **messages** — id, conversation_id, sender (user/kd/admin), channel (web/email/whatsapp), content, intent_label, kb_sources (json), tokens, created_at.
- **kb_documents** — id, agency_id, title, source_type (upload/crawl/conversation), source_url/file_path, language, version, status (draft/approved/archived), last_verified_at, timestamps.
- **kb_chunks** — id, kb_document_id, agency_id, chunk_text, embedding_ref (vector id), metadata (json).
- **crawl_jobs** — id, agency_id, name, seed_urls (json), include/exclude patterns (json), depth, max_pages, file_types (json), schedule_cron, is_enabled, review_required, last_run_at, last_status.
- **crawl_runs** — id, crawl_job_id, started_at, finished_at, pages_fetched, added, updated, skipped, errors (json), log_ref.
- **otp_verifications** — id, target (email/phone), code_hash, channel, expires_at, attempts, verified_at.
- **settings** — key, value (json), group, updated_by, updated_at [**Corrad: reuse**].
- **audit_logs** — id, actor_id, action, entity, entity_id, before (json), after (json), ip, created_at.
- **notifications** — id, type, channel, to, template, payload (json), status, sent_at.

Key relationships: an agency has many KB documents, conversations, crawl jobs, and agency-scoped admins; a conversation has many messages; a KB document has many chunks.

12. API surface (indicative)

RESTful Laravel API consumed by the Vue 3 front end. All admin endpoints require auth + RBAC + agency scoping.

Public / chat

- POST /api/chat/session — start a session (language).
- POST /api/chat/message — send a user message, returns KD's grounded reply + intent + suggested actions.
- POST /api/chat/identity — submit name/email/phone + consent.
- POST /api/chat/verify/request — request email/phone OTP.
- POST /api/chat/verify/confirm — confirm OTP.
- POST /api/chat/feedback — rate an answer.
- POST /api/chat/escalate — request a human; queues conversation.

Admin

- .../auth/* — login, logout, 2FA, password reset **[Corrad]**.
- .../users, .../roles — user & RBAC management **[Corrad]**.
- .../agencies — agency CRUD; .../agencies/{id}/admins — agency sub-admins.
- .../kb/documents, .../kb/documents/{id}/reindex, .../kb/review-queue — KB management.
- .../conversations, .../conversations/{id}, .../conversations/{id}/reply (email/whatsapp), .../conversations/{id}/assign.
- .../crawl/jobs, .../crawl/jobs/{id}/run, .../crawl/runs — crawler config, manual run, logs.
- .../dashboard/metrics — role-scoped metrics.
- .../settings — read/update settings groups **[Corrad]**.
- .../audit-logs — read audit trail.

Async workers / scheduled

- Crawl scheduler (per crawl_jobs.schedule_cron).
- Ingestion + embedding workers.
- Notification (email/SMS/WhatsApp) workers.
- Conversation-to-KB redaction & review worker.

13. Suggested build phases (for the developer)

1. **Foundation** — clone/adopt Corrad-Laravel; confirm auth, RBAC, settings, media, Vue 3 shell; seed roles (super_admin/admin), the three agencies, and a demo super-admin.
2. **Public shell** — build the MOHR front-page replica (FR-PUB-01/02) and the KD floating widget (FR-PUB-04/05).
3. **Onboarding flow** — language pick, intro, three-agency mention, name/email/phone capture + OTP verification + consent (FR-PUB-07...14).
4. **KB + RAG core** — ingestion pipeline, embeddings, retrieval, LLM answering with persona/guardrails (Sections 6 & 9); wire KD answering to it (FR-PUB-15...21).
5. **Admin modules** — KB management, conversation inbox + email/WhatsApp reply, agency admin, dashboards (Section 5).
6. **Crawler** — configurable, scheduled crawling into the KB with review/logs (Section 5.7).
7. **Conversation-to-KB loop** — redaction, review queue, approval (FR-KB-09, FR-ADM-12).
8. **Hardening** — security, PDPA, accessibility, performance, tests, CI, seed/demo data.

14. Acceptance criteria (sampling)

- **AC-1** — Opening KD prompts for BM/EN/ZH/TA; selecting Tamil makes all subsequent KD copy Tamil. *(FR-PUB-07)*
- **AC-2** — KD introduces itself and names the three focus agencies as quick-reply options. *(FR-PUB-08/09)*
- **AC-3** — Providing an email triggers an OTP; entering the correct code marks the email verified; the same for phone via SMS/WhatsApp. *(FR-PUB-11/12)*
- **AC-4** — Declining to give a name still lets the user ask and receive a grounded answer. *(FR-PUB-14/15)*
- **AC-5** — A job-search question is classified and answered from the HRDCorp/MyFutureJob KB with a deep link to the portal. *(FR-PUB-16/17)*
- **AC-6** — Asking something absent from the KB makes KD say it doesn't have that information and offer human escalation rather than inventing an answer. *(FR-PUB-18, FR-AI-04)*
- **AC-7** — An agency admin sees only their agency's conversations and KB, and can add a sub-admin for that agency. *(FR-ADM-03/04/19)*
- **AC-8** — An admin opens a conversation, reads the full log, and replies by email or WhatsApp; the reply appears in the transcript. *(FR-ADM-14/15)*
- **AC-9** — Super Admin adds a fourth agency with its own KB and admins without any code change. *(FR-ADM-18, NFR-05)*
- **AC-10** — A crawl job set to "daily 02:00" runs on schedule, ingests changed MOHR pages into the correct agency KB, and logs the run. *(FR-ADM-23...27)*
- **AC-11** — On login, an agency admin lands on a dashboard scoped to their agency's metrics. *(FR-ADM-20)*
- **AC-12** — A reviewed past conversation is approved and becomes retrievable KB content; PII is redacted. *(FR-KB-09, FR-ADM-12)*
- **AC-13** — Changing the KD avatar/name/welcome copy in Settings updates the public widget without code changes. *(FR-SET-02, FR-BRD-01)*

15. Open items to confirm with MOHR

- Official "**Dewi**" **backronym** wording and the final approved KD avatar artwork.
- Availability and specs of **agency lookup APIs** (claim/complaint/job status) vs guided-link-only for phase 1.
- Chosen **LLM provider/model** and **embedding model**, and whether hosting must remain in-country.
- **WhatsApp Business** provider and sender number; SMS fallback.
- **Data-retention** period for transcripts and PII, and the erasure-request process.
- Hosting target (government cloud vs commercial cloud, region).

End of SRS v1.0.